

Working with strings

Coders use the word “string” for any data made up of a sequence of letters or other characters. Words and sentences are stored as strings. Almost all programs use strings at some point. Every character that you can type on your keyboard, and even those you can’t, can be stored in a string.

1 Strings in variables

Strings can be put into variables. Type this code into the shell window. It assigns the string 'Ally Alien' to the variable **name** and then displays it. Strings must always have quotation marks at the beginning and end.

2 Combining strings

Variables become really useful when you combine them to make new variables. If you add two strings together, you can store the combination in a new variable. Try this out.

EXPERT TIPS

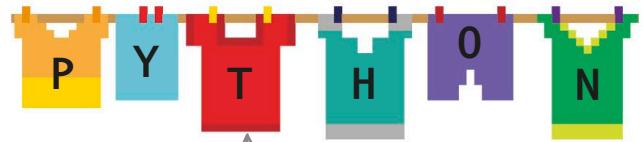
Length of a string

You can use a handy trick, **len()**, to count the number of characters in a string (including the spaces). The command **len()** is an example of what coders call a function. (You’ll use lots of functions in this book.) To find out how many characters there are in 'Welcome to Earth, Ally Alien', type the line below into the shell once you’ve created the string, then hit enter/return.

```
>>> len(message)
```

```
28
```

The number of characters counted



A string is simply a sequence of characters.

The quote marks show that the variable contains a string.

```
>>> name = 'Ally Alien'
>>> print(name)
Ally Alien
```

Hit the enter/return key to print the string.

Remember the quote marks.

```
>>> name = 'Ally Alien'
>>> greeting = 'Welcome to Earth, '
>>> message = greeting + name
>>> print(message)
Welcome to Earth, Ally Alien
```

The + symbol joins one string to another.

The quote marks aren't shown when you print a string.

